1. what are 3 requirements for a C function that
is used as a FreeRTOS task

-returns void
-takes a void* as a parameter
-must not hit a return statement

2. Figure 11 shows an execution pattern for two tasks.
what causes task 1 and task 2 to alternate execution

The alternation occurs because both tasks have equal
priority and repeatedly yield or block, so the scheduler
time-slices (round-robins) between them.

3. why is the configTICK_RATE_HZ typically set to 100.
what reasons are there to set it to 10 or 1000?

it is a good balance of resolution while minimizing CPU
overhead from constant tick interrupts and switching.

if it was faster it would spend a lot of time on the OS
rather than the tasks. If it was slower than it would
probably fail a task since it took too long to switch tasks.

4. using figure 15. name scenarios for all 8 state transitions

- suspend->ready. another task calls vTaskResume using the
task handle
- ready->running. the scheduler at a tick selects this task to
be running.
- running->ready. the scheduler switches task but task is still
in the schedule(not suspended)
- running->block. the task requests a resource or a delay and
cant continue.
- blocked->ready. the blocking action has been solved and task
is switched.
- ready->suspended. another task has called the vTaskSuspend
and the task is no longer ready.
- running->suspended. the task itself has calls VTaskSuspend on
itself
- blocked->Suspended. a blocked task is forcibly suspended by
another task

5. what is the purpose of the FreeRTOS Idle Task

- so the scheduler has something safe to run if there are
no current tasks.